J Chron Dis 1976, Vol. 29, pp. 175–188. Pergamon Press. Printed in Great Britain

THE COMBINATION OF RANDOMIZED AND HISTORICAL CONTROLS IN CLINICAL TRIALS

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(*Received in revised form* 17 *June* 1975)

INTRODUCTION

THE CONCEPT of treatment randomization has been an important aspect of the design of scientific experiments ever since the pioneering work of Fisher [1]. However, the introduction of randomization to clinical trials did not occur until the 1950's with the studies of Bradford Hill [2]. Today, many clinical researchers accept the idea that the proper evaluation of a new therapy in a clinical trial requires the inclusion of a *randomized control* group of patients treated with a standard therapy. However, if one scans the pages of any current medical journal it is evident that a substantial element of clinical research continues to ignore this aspect of good scientific experimentation.

Often the published results on a new therapy may have *no* control group, leaving the author (and reader) to make his own somewhat arbitrary assessment of the merits of the new therapy in relation to other more standard therapies. This type of uncontrolled research can still be a valid contribution, especially as a preliminary evaluation of a new drug with regard to toxicity and the determination of an acceptable dosage, but unfortunately there is an abundance of therapies accepted as standard on the basis of such uncontrolled results. The problems arising from this approach are evident in such long-standing practices as radical mastectomy for breast cancer and oral agents for diabetes, both of which are currently a cause of much controversy because of the lack of past evidence in a controlled study to establish their efficacy.

An alternative to randomized controls is the use of retrospective data on a standard treatment, that is a *historical control* group. This has the advantage that *all* patients in the trial will receive the new treatment which the investigators believe, or hope, to be superior. The major problem with historical controls is that one is unable to ensure comparability between the groups of patients and methods of evaluation for the new and standard treatment. Obvious bias may occur if the investigator is selective in the patients assigned to a new treatment but a more subtle unconscious bias in patient selection may arise because of unforeseen differences in time and place. Also, patients entering the trial for a new therapy may receive superior medical care and supervision which contributes to the 'placebo effect' that mere entry into a clinical trial improves the patient's ability to respond. Furthermore, inconsistency in the evaluation of the effect of therapy, changes in the quality of patient, and variation between investigators can seriously interfere with any historical comparison.

Therefore, as discussed by Chalmers *et al.* [3] and Ingelfinger [4] the case for randomized controls is convincing although Gehan and Freireich [5] argue that a place still exists for a careful choice of historical controls. The National Cancer Institute [6] provides an extensive discussion of controls in cancer clinical trials. One common and possibly erroneous assumption when comparing the relative merits of randomized and historical controls is that one has to make a clear choice between the two. It seems sensible to propose that a degree of compromise is possible whereby the comparative evaluation of a new treatment uses *both* randomized and historical controls. In practice, such a combined use of controls frequently occurs in a subjective manner, since the results of any randomized clinical study are considered in the context of past results. Such comparisons with previous results seem particularly valuable if made within the same research center, but all too often a more arbitrary comparison is made with results obtained from other sources.

For example, suppose in a randomized study a new treatment for a disease has a 40% response rate compared with 20% on the standard treatment, but that the response rate on the standard treatment in a previous study was 35%. Those researchers arguing exclusively for randomized controls would claim a 20% improvement on the new treatment but when the trial results are placed in a broader historical setting one would have to argue more cautiously.

The main problem with this subjective interpretation of historical controls with a randomized trial is that the researcher decides *post hoc* whether to place any value in the historical data and he decides arbitrarily *how much* this data should influence his conclusions. Thus, the approach has been qualitative rather than quantitative. The objective of this paper is to consider the combination of randomized and historical controls in a more quantitative manner so that formal methods of statistical inference can be applied.

Besides influencing the analysis of a randomized trial, the use of historical data should influence its design. For instance, suppose one wishes to compare a new treatment against standard therapy. The usual procedure in a randomized trial is to assign half the patients to each treatment and a variety of randomization procedures are available [7]. However if historical controls on the standard therapy are available this 50/50 split seems less than optimal. Therefore, this paper also considers the assignment of a smaller proportion of patients to a randomized control when '*acceptable*' historical controls are available.

ACCEPTABLE HISTORICAL CONTROLS

As a preliminary to this section, it seems appropriate to include the conclusion of Ingelfinger that "...ethical, as well as scientific, considerations require that medicine depend on the most reliable and the best controlled data available—the kind of data that is sought by randomized clinical study." The author is in general agreement with this statement but also feels that the additional presence of acceptable historical data cannot be ignored in the full comparative evaluation of a new treatment. The acceptability of a historical control group requires that it meets the following conditions:

1. Such a group must have received a precisely defined standard treatment which must be the same as the treatment for the randomized controls.

2. The group must have been part of a recent clinical study which contained the same requirements for patient eligibility.

3. The methods of treatment evaluation must be the same.

4. The distributions of important patient characteristics in the group should be comparable with those in the new trial.

5. The previous study must have been performed in the same organization with largely the same clinical investigators.

6. There must be no other indications leading one to expect differing results between the randomized and historical controls. For instance, more rapid accrual on the new study might lead one to suspect less enthusiastic participation of investigators in the previous study so that the process of patient selection may have been different.

Only if all these conditions are met can one safely use the historical controls as part of a randomized trial. Otherwise, the risk of a substantial bias occurring in treatment comparisons cannot be ignored. For instance, 'literature' controls as referred to in [5] are unsuitable since in general most of the six conditions are not met.

With the increasing acceptance of randomized trials for the evaluation of new treatments, the occurrence of such acceptable historical data is also increasing. Thus, in many clinical research centers each trial is not an isolated event but is one of a sequence of such trials for a particular disease. For instance, in multicenter co-operative groups in cancer trials it is common practice to activate one randomized trial for a disease as soon as the previous one is terminated. Furthermore, one treatment in the previous trial will usually be the standard control in the next trial.

One example of such an organization is the Eastern Co-operative Oncology Group (ECOG) and Table 1 gives a few illustrations of such sequences of chemotherapy studies in advanced cancer. In each case the best treatment in the previous study was continued as a randomized control treatment in the current study. The presence of this randomized control was considered essential for the scientific validity of each study, since otherwise one would have no check on investigator bias as regards patient selection for the new study. In fact, it tends to eliminate such bias. ECOG is one of many such groups supported by the National Cancer Institute and it is their general policy that purely historical controls are unsatisfactory and randomization is an essential requirement of all such Phase III (comparative) studies.

However, due to a lack of any theoretical justification, there has been some inconsistency in deciding what proportion of patients in the new study should receive the standard treatment. Thus, the current melanoma and sarcoma studies in Table 1 have ignored the presence of historical controls by assigning equal

TABLE 1. SEQUENCES OF STUDIES IN THE EASTERN CO-OPERATIVE ONCOLOGY GROUP

Disease site	Previous study	Current study	Date current study activated
Breast	Cytoxan + Methotrexate + 5-FU v Melphalan	→CMF continued v CMF + Prednisone v Adriamycin + Vincristine	October 1973
Lung (squamous and large cell)	Cytoxan v Cytoxan + CCNU	→ Cytoxan continued v Adriamycin	April 1973
Melanoma	DTIC v TIC Mustard	→ DTIC continued v Methyl CCNU v DTIC + Methyl CCNU	December 1972
Sarcoma	Adriamycin v Cycloleucine	→ Adriamycin continued v Vincristine + Cytoxan + Adriamycin	August 1974

numbers of patients to all treatments. The two-arm lung study is assigning 1/3 of the patients to the control and the three-arm breast study will have 1/4 of patients on the control arm. The results in the following section should help in deciding on the adjustment in proportion entered as randomized controls in the presence of historical controls. Also, tied in with such design aspects are the methods of statistical analysis and inference applicable to the combination of randomized and historical controls.

THE STATISTICAL MODEL

Suppose one performs a clinical trial to compare the effect of a new treatment with a standard control treatment. Randomization is used to assign patients to treatments, though not necessarily in equal proportions, and there also exist acceptable historical data on the control treatment.

The groups of patients on the new treatment, the randomized control and the historical control are referred to as T, R and H respectively. Let N_T, N_R and N_H be their respective sample sizes and let D_T, D_R and D_H denote the data on treatment evaluation for each group.

The basis questions to be answered are:

(1) How can one combine D_R and D_H, the two sets of control data, for comparison with D_T?

(2) In planning the trial, how should one determine sample sizes N_T and N_R given the historical data D_H based on a sample size N_H?

It would seem impossible to provide a completely general solution to these problems given the variety and complexity of data one may use in the evaluation of treatment. Therefore, a more specific model is considered, the solution for which should provide some insight into the more general problem.

Suppose that the evaluation of treatment for a patient can be summarized by a single quantitative measurement X. For example, in trials for advanced cancer X might be the patient's survival time or an ordered scale for the assessment of objective tumor response. For any patient in a particular treatment group the value x of X can be thought of as an observation from a random variable X for which the mean and variance are generally unknown. Furthermore, let this random variable have a normal distribution. This assumption is not unduly restrictive since with reasonably large sample sizes the central limit theorem ensures the adequacy of normal theory. Examples of this fact for binomial and exponential data are shown below. Therefore, let the patient groups T, R and H have associated normal random variables for the measurement X with unknown means μ_T, μ_R and μ_H and variances σ_T^2, σ_R^2 and σ_H^2. The obervable sample means of X for these groups may be denoted by $\bar{x}_T$, $\bar{x}_R$ and $\bar{x}_H$ respectively where $\bar{x}_T = \Sigma_T x/N_T$ etc.

The main objective of the trial may be considered the accurate estimation of $\mu_T - \mu_R$, a quantitative measure of the relative effectiveness of the new and standard treatments. In a randomized trial the usual point estimate of this quantity is $\bar{x}_T - \bar{x}_R$ and in the absence of historical data this is the best estimate in that it is unbiased and has minimal standard error.

However, the presence of historical data summarized by $\bar{x}_H$ cannot be ignored if one wishes to obtain the optimal estimator for $\mu_T - \mu_R$. The major problem to be overcome is that μ_R and μ_H are not necessarily equal, even if the conditions for an acceptable historical control in section 2 are met. One must allow for the possibility that there exists an *unknown bias* $\delta = \mu_R - \mu_H$ in the historical control. Even if the historical controls are 'acceptable' as defined in the previous section, one must still accept them as potentially biased, or unreliable, for one or more of the reasons described in the introduction. In fact, the use of randomized controls is based on the probable existence of such a non-zero bias. In general, one is in a state of semi-ignorance as to the value of δ: one is unable to assess whether it is positive or negative but at the same time one may have some idea as to the range of possible values. One suitable way out of this dilemma is to treat δ as a random variable, setting its mean equal to zero to indicate that the direction of bias is unknown and its variance equal to some fixed quantity σ_δ^2. Also, it is convenient to suppose δ has a normal distribution which seems a reasonable choice. This Bayesian prior distribution for δ expresses the degree of confidence one has in the historical data. The choice of σ_δ is difficult but as a general guideline the author feels one should err on the side of caution by assigning a reasonably large value to σ_δ. That is, it is better to place too little rather than too much confidence in historical data. Perhaps in practice one should consider several possible values for σ_δ in order to assess the effects this has on experimental design and analysis. Better still, one might be able to use two or more sets of historical data to assess more objectively the value of σ_δ; see below for details.

THE COMBINATION OF CONTROLS IN ANALYSIS

$\bar{x}_R$ and $\bar{x}_H$ can both be considered estimators of μ_R and the purpose of this section is to determine a combined estimator, $\bar{x}_c$ say. This is achieved using Bayesian statistical methods.

Prior to the historical data one may suppose complete ignorance regarding the value of μ_H (i.e. a uniform prior distribution). The historical data then lead to μ_H having a normal posterior distribution mean $\bar{x}_H$ and variance σ_H^2/N_H. Now, $\mu_R = \mu_H + \delta$, the sum of two normal random variables. Thus, after incorporating historical data but prior to the inclusion of randomized controls, μ_R has a normal distribution mean $\bar{x}_H$ and variance $\sigma_H^2/N_H + \sigma_\delta^2$. This can be considered the prior distribution of μ_R about to be modified by the subsequent data from the randomized controls mean $\bar{x}_R$ and variance σ_R^2. From a simple application of Bayes theorem, the posterior distribution of μ_R is normal

with mean
$$\frac{[(\sigma_H^2/N_H) + \sigma_\delta^2]\bar{x}_R + (\sigma_R^2/N_R)\bar{x}_H}{(\sigma_R^2/N_R) + (\sigma_H^2/N_H) + \sigma_\delta^2} = \bar{x}_c, \tag{1}$$

and variance
$$\left[(N_R/\sigma_R^2) + \frac{1}{\sigma_\delta^2 + (\sigma_H^2/N_H)}\right]^{-1} = V_c. \tag{2}$$

Formula (1) can be rewritten as

$$\bar{x}_c = (\bar{x}_R + W\bar{x}_H)/(1 + W)$$

where $W = (\sigma_R^2/N_R)/(\sigma_H^2/N_H + \sigma_\delta^2)$. Thus, the optimal point estimate $\bar{x}_c$ for μ_R is a weighted sum of the sample means $\bar{x}_R$ and $\bar{x}_H$.

The following discussions center on the practical usefulness of this model and the result as applied to actual clinical trials:

(a) *Statistical inference*

Having determined $\bar{x}_c$ from equation (1), one is interested in making inferences about the difference between μ_T and μ_R in order to assess the relative effectiveness of the new and control treatments. Because of the manner in which the model is formulated it is natural to use Bayesian methods of inference. Thus, if one assumes a uniform prior for μ_T the posterior distribution of μ_T is normal with mean $\bar{x}_T$ and variance σ_T^2/N_T. Hence, the posterior distribution of $\mu_T - \mu_R$ is normal with mean $\bar{x}_T - \bar{x}_c$ and variance $\sigma_T^2/N_T + V_c$ where $\bar{x}_c$ and V_c are defined in formulae (1) and (2). It is now an easy matter to make inferences about $\mu_T - \mu_R$ as is shown in the example below. There is a multitude of more complex inference problems one could examine to incorporate non-normality and/or stratification but the basic principles of the model should remain applicable. The asymptotic normality of any sample mean should make normal theory adequate for most data sets; alternatively transformation of the variable may help, as in section (d. ii). One should remember when applying these methods that the resulting inferences depend not only on the observed data but also on one's subjective determination of a value for σ_δ^2, the variance of the historical bias.

(b) *Estimation of variances*

In general, the variances σ_T^2, σ_R^2 and σ_H^2 are unknown and the simplest solution is to substitute sample variances S_T^2, S_R^2 and S_H^2 where $S_T^2 = \Sigma_T\,(x - \bar{x}_T)^2/(N_T - 1)$

etc. In many trials it is safe to assume that the variance is the same for all treatment groups in which case the appropriate estimated variance is

$$[(N_T - 1)S_T^2 + (N_R - 1)S_R^2 + (N_H - 1)S_H^2]/(N_T + N_R + N_H - 3).$$

In the instance of unknown variances the posterior distributions depend on Student's t, but this would seem an unnecessary elaboration for sample sizes N_H etc. above 20.

(c) *Stratified treatment groups*

So far, by letting X be a simple random variable in each treatment group the method has assumed that a homogeneous population of patients is being treated. This is equivalent to ignoring the effect of important patient characteristics on the response to treatment. For instance, in advanced breast cancer, patients who are not ambulatory, have short free period, and have liver metastases have a dire prognosis. In both the design and analysis of a trial it is important to allow for such patient characteristics using stratification procedures [7]. Thus, the patient population can be divided into several strata and the results of this section applied separately for each stratum. The estimation of variances and methods of statistical inference may need to combine results from different strata but this should present no problems.

(d) *Other types of data*

(i) *Binary data.* In many trials it is impossible to obtain a single quantitative measure of response to treatment for a patient. Instead, a set of rules may be defined to determine if each patient achieved a response or not. For example, the criteria for objective tumor response in solid cancer chemotherapy trials in ECOG include a variety of requirements for tumor shrinkage. The response rate for a treatment is the proportion of patients who responded according to the definition and this can be thought of as the mean of a binary variable scoring 1 for response and 0 for no response. Thus, the results of this section can be applied to such binary data. The appropriate variance estimator for treatment T say is $\bar{x}_T(1 - \bar{x}_T)/N_T$ where $\bar{x}_T$ is now the proportion of patients responding on treatment T.

(ii) *Exponential data.* In many actual trials the sample observations follow an exponential distribution, a common example being the survival times of patients with advanced cancer. The exponential distribution function is $1 - \exp(x/m)$ where m is the mean survival time. For any sample of patients, m can be estimated by the observed mean survival time y = sum of all survival times for patients both dead and alive/number of deaths. Now, it can be shown that log y has an asymptotic normal distribution with mean = log m and variance = $1/N$ where N is the number of deaths. This enables one to apply the normal theory described earlier to exponential survival data from randomized and historical controls, R and H respectively.

Let m_R, m_H be the true (unknown) mean survival times, N_R, N_H the observed numbers of deaths and y_R, y_H the observed mean survival times. Let log $(m_R/m_H) = \delta$, the historical bias random variable with a normal distribution mean

0, variance σ_δ^2. This is equivalent to m_R/m_H having a log-normal distribution with median 1 and variance

$$\exp \sigma_\delta^2 (\exp \sigma_\delta^2 - 1) \simeq \sigma_\delta^2 \text{ for small values of } \sigma_\delta.$$

Now, assuming a uniform prior distribution for $\log m_H$ it follows that the posterior distribution of $\log m_H$ is asymptotically normal with mean $\log y_H$ and variance $1/N_H$. Hence, using the same principles that led to formulae (1) and (2) it is easy to show that the posterior distribution of $\log m_R$ is asymptotically normal with mean

$$\frac{[(1/N_H) + \sigma_\delta^2] \log y_R + (1/N_R) \log y_H}{(1/N_R) + (1/N_H) + \sigma_\delta^2} = \log y_c, \text{ say} \tag{3}$$

and variance

$$[N_R + N_H/(1 + N_H\sigma_\delta^2)]^{-1}. \tag{4}$$

By taking the exponential of formula (3) one obtains y_c, a point estimate of m_R the mean survival time for the randomized control treatment. Lastly, for the new treatment T the posterior distribution of $\log m_T$ is normal with mean $\log y_T$ and variance $1/N_T$. One can now draw inferences about the comparability of m_R and m_T based on these posterior distributions.

Example 1

In 1972 the ECOG ran a randomized trial for patients with advanced melanoma comparing two chemotherapeutic regimens, DTIC and TIC-Mustard. One major objective of the study was to compare the survival experience of patients on the two treatments. DTIC was also a treatment in the previous ECOG melanoma trial and furthermore this historical control group appeared to meet the acceptability criteria defined earlier so that the combination of randomized and historical controls can be carried out as defined above.

Firstly, the survival experience for each treatment group is summarized in Table 2. Also, an exponential model seemed reasonable according to survival plots of the data. Now, the historical bias $\delta = \log(m_R/m_H)$ has been assigned a variance $\sigma_\delta^2 = 0{\cdot}01$ which corresponds to m_R/m_H having S.D. $\sqrt{e^{0.01}(e^{0.01} - 1)} = 0.1$. This is equivalent to saying that the potential historical bias in the mean survival times may be of the order of 10%.

Then, applying formulae (3) and (4) the posterior distribution of log mean survival for the randomized controls DTIC is approximately normal with mean $\log y_c = 3.28$ and variance 0.0112. Thus, $y_c = 26.5$ weeks is the combined point estimate of mean survival on DTIC. For the new treatment TIC-Mustard the log

TABLE 2. SURVIVAL RESULTS FROM TWO MELANOMA TRIALS

	Treatment	No. of Patients	No. of Deaths (N)	Estimated mean survival time (y)
1972 Trial	TIC-Mustard (T)	70	49	27.1 weeks
	DTIC (R)	80	59	28.4 weeks
Previous trial	DTIC (H)	57	44	23.2 weeks

mean survival has an approximate normal posterior distribution with mean 3.30 and variance 0.0204. Thus, log (m_R/m_T = the ratio of true mean survival times for DTIC and TIC-Mustard) has an approximate normal posterior distribution with mean = 3.28 − 3.30 = −0.02 and S.D. = 0.0112 + 0.0204 = 0.18. This leads to a normal deviate of −0.02/0.18 = −0.1 which makes it quite clear that there is no evidence of a survival difference between treatments.

Now, suppose one wanted to increase the potential magnitude of the historical bias. For instance, if $\sigma_\delta^2 = 0.04$ then m_R/m_H has S.D. 0.2, double the previous value. Then, y_c the point estimate for mean survival on DTIC = 27.2 weeks which is closer to the observed value 28.4 weeks for randomized controls alone. Thus, by increasing the bias variance σ_δ^2 one places greater emphasis on the randomized controls and the historical data becomes of less value.

(e) *Estimating the potential bias*

The application of the above model is quite sensitive to the potential magnitude of the historical bias as represented by the variance σ_δ^2. In general, the very nature of the problem makes it impossible to assign an accurate, objectively determined value to σ_δ^2 and instead one usually has to make a subjective decision regarding what is a reasonable value. In fact, one may wish to repeat the analysis with several alternative values, thus expressing varying degrees of trust in the historical controls.

However, if several groups of acceptable historical controls are available, say $K \geqslant 2$, one may be able to obtain an estimate of σ_δ^2 based on a random-effects model for the one-way analysis of variance, see [9] for details. σ_δ^2 would be the component of variance between groups which for equal sample sizes is estimated by (between group mean square—within group mean square)/N of observations per group. For small values of K, say under 5, this estimate would not be particularly accurate but could still be a rough guide. Larger values of K are perhaps most likely to occur in animal trials. For instance, in the ECOG at most three groups of controls existed for any disease site. However, by pooling estimates of σ_δ^2 from several disease sites, clinical trial groups such as the ECOG may eventually be able to obtain a more accurate estimate of σ_δ^2 acceptable for use in a wide range of future trials.

(f) *An alternative approach*

Faced with both randomized controls and acceptable historical controls for a new treatment most researchers would choose one of the following alternative methods of analysis:

(i) They would use only the randomized controls because *either* they view any historical data as being unreliable *or* there is a suggestion of a difference in the two sets of control data.

(ii) They would simply pool the two sets of control data because they feel the historical data are reliable enough not to require any separate consideration *and also* the two sets of data look similar.

The main failing in this approach is that the choice between (i) and (ii) is somewhat arbitrary. However, one could devise a more objective scheme for selecting (i) or (ii). Possibilities would be to select (ii), the pooling of control data, if (a) the difference between $\bar{x}_H$ and $\bar{x}_R$ is below some pre-specified limit, or (b) a significance test on the equality of population means, μ_H and μ_R, results in a significance level greater than some pre-specified value. Such a procedure is attempting to assess the magnitude of the bias $\delta = \mu_R - \mu_H$ referred to earlier. However, if the historical controls are acceptable according to the previously defined criteria the bias is likely to be small and undetectable in the data. Consequently, occasional rejection of historical data may occur more because of random variability between samples rather than because of substantial bias. Thus, it seems better to incorporate unknown bias into a statistical model such as that described earlier in this section rather than make any attempt to estimate bias.

SAMPLE SIZE FOR A RANDOMIZED CONTROL GROUP

Traditionally, the design of a randomized clinical trial requires *equal* numbers of patients on each treatment. If historical controls are absent (or ignored) this approach usually provides the most accurate comparison of treatments. For instance, using the notation introduced earlier, setting $N_T = N_R$ results in $\bar{x}_T - \bar{x}_R$ having minimum variance provided $\sigma_T = \sigma_R$.

However, as is shown in this section, the use of historical controls influences the design of a randomized trial by enabling a smaller proportion of patients to be in the randomized controls, i.e. N_R may be made less than N_T.

As in the previous section on analysis, a completely general solution for sample size of randomized controls in the presence of acceptable historical data would seem impossible and therefore attention is restricted to the more specific situation defined earlier in the statistical model and the same notation is now used.

Suppose the total sample size for the randomized trial $N = N_T + N_R$ has already been determined. The problem is to determine an 'optimal' value for N_R given the values N and N_H. In addition one needs to have values for σ_δ^2, the bias of the historical control group, and σ_H^2, σ_R^2 and σ_T^2 the variances of X for the three treatment groups. The determination of reliable values or estimates for these variances has already been discussed. In this situation, prior to the randomized trial, one does not have sample variances S_T^2 and S_R^2 in which case it seems appropriate to set the three variances σ_H^2, σ_R^2 and σ_T^2 all equal to S_H^2, the sample variance of the historical data.

The 'optimal' value of N_R is to be determined by consideration of an intended statistical analysis of the type described above. That is, one intends to obtain a normal posterior distribution for $\mu_T - \mu_R$ the true difference in treatment means which has mean $\bar{x}_T - \bar{x}_c$ and variance $\sigma_T^2/N_T + V_c$ where $\bar{x}_c$ and V_c are defined by formulae (1) and (2). One needs to choose N_R such that this variance is minimized—that is one wishes to design the new trial so that $\mu_T - \mu_R$ is estimated as precisely as possible under the constraint that $N_T + N_R = N$ is fixed.

Now, this variance $\sigma_T^2/N_T + V_c$ can be rewritten using formula (1) as

$$\frac{\sigma_T^2}{N - N_R} + \frac{\sigma_R^2[(\sigma_H^2/N_H) + \sigma_\delta^2]}{\sigma_R^2 + N_R[(\sigma_H^2/N_H) + \sigma_\delta^2]}. \tag{5}$$

By setting the first derivative of (5) with respect to N_R equal to zero, one obtains the 'optimal' value of N_R which minimizes the variance of $\mu_T - \mu_R$.

This optimal value is

$$N_R = \frac{\sigma_R}{\sigma_R + \sigma_T}\left(N - \frac{\sigma_R \sigma_T}{(\sigma_H^2/N_H) + \sigma_\delta^2}\right). \tag{6}$$

If $\sigma_H = \sigma_R = \sigma_T = \sigma$, then the formula simplifies to

$$N_R = \tfrac{1}{2}\left(N - \frac{N_H}{1 + (\sigma_\delta^2 N_H/\sigma^2)}\right). \tag{7}$$

Before the above procedure is illustrated with an example it seems appropriate to discuss a few practical implications:

(a) *More complex situations*

The above model was necessarily made simple in order that a reasonably general and uncomplicated procedure could be described for the determination of N_R. In a real-life sequence of clinical trials the situation may be more complex:

(i) Problems concerning the stratification of treatment groups, estimation of variances and use of binary data have already been discussed.

(ii) There may be several new treatments to be compared with the control. Provided the main objective is to compare each individual new treatment with the control, the above formulae need only minor modification.

(iii) In many clinical trials, each patient has several response variables measured for the purpose of comparing treatments. For instance, in a trial for patients with advanced breast cancer both maximum tumor shrinkage and patient survival are used to assess response to treatment. In such cases, the sample size N_R for the randomized controls may be determined separately for each response variable and the eventual choice of N_R could be a weighted mean of these values, weights being according to relative 'importance' of the response variable as decided by the investigator.

(b) *Planning the randomization*

Having chosen N and N_R, the randomization requires that a patient enters the control group with probability N_R/N. This is a simple matter to set up if one uses purely random sampling, but careful planning is needed if one uses any form of restricted randomization. For example, with a stratified randomization scheme one usually arranges to have permuted blocks of treatment assignments within each stratum. It would then be convenient to have N_R/N rounded off to some simple fraction (e.g. 1/3 or 2/5) so that small permuted blocks could be formed (e.g. of length 3 for $N_R/N = 1/3$ and 5 for $N_R/N = 2/5$).

(c) *Alternative procedures*

The determination of sample sizes in a clinical trial is usually based on three factors: statistical requirements, available accrual and cost. The standard statistical approach is to determine the sample size required to detect with some high probability a specified population mean treatment difference, $\mu_T - \mu_R$, at a specified level of statistical significance. Such procedures are described in many sources, for example [8]. However, from the author's experience the available accrual and economic considerations often take priority in the determination of sample size. For instance, one may first calculate the sample size obtained from 2 yr accrual in a certain institution and only afterwards determine the statistical power this provides for detecting treatment differences. Thus, the decision about the total sample size N is usually a subjective combination of the objective knowledge of statistical power, accural rate and cost.

Therefore, the assessment of N_R, the sample size for randomized controls in the presence of historical controls, is based upon the assumption that N, the total number of patients in the trial, has already been determined. An alternative procedure would have been to determine both N and N_R after specifying the required statistical power. This has not been described here since the author feels the procedure described in detail above would be of more general application.

(d) *Sequential designs*

This section has so far dealt with the calculation of a fixed value N_R given a fixed total sample size N. In practice, many clinical trials are run in a sequential manner, accrual being stopped once a satisfactory conclusion can be made from available data. The stopping rule for such a trial should be made according to statistical definitions made in the initial design, but in all too many instances the stopping rule is ill-defined and subjective.

The combination of historical and randomized controls in a sequential design for a clinical trial has not been explored in this paper, but could be a useful area for future research.

The following example is drawn from an actual sequence of clinical trials in the Eastern Cooperative Oncology Group.

Example 2

A randomized trial for advanced small cell lung cancer patients compared Cytoxan alone with Cytoxan + CCNU. The latter treatment was distinctly superior so the next trial is to be a comparison of Cytoxan + CCNU with Cytoxan + CCNU + Procarbaxine. In each trial there are two main end-points, objective tumor response and survival. In the first trial there were 88 patients treated with Cytoxan + CCNU. The tumor response rate was 37/83 = 45%, there being 5 non-evaluable cases. 65/88 patients have died so far and the estimated mean survival = 27.8 weeks using an exponential survival distribution. Now, it is proposed that 200 patients be entered on the new trial. The question to be asked is 'How many of these 200 patients should be entered on the randomized control, Cytoxan + CCNU, for which there are already 88 patients in a historical control group?'.

Separate solutions will be obtained using tumor response and survival data:

(a) *Tumor response*. Using earlier notation, $N_H = 83$ and $N = 200$. $\bar{x}_H = 45\%$ and from the binomial distribution σ_H is estimated to be $\sqrt{45 \times (100 - 45)} = 49.7\%$. It is assumed that $\sigma_T = \sigma_R = \sigma_H$.

Suppose σ_δ, the standard deviation of the bias in the historical controls, is set equal to 3%. Then, applying formula (7) we obtain

$$N_R = \tfrac{1}{2}\left(200 - \frac{83}{1 + [(3/49.7)]^2\, 83}\right) = 68.1.$$

(b) *Survival.* The survival of patients with advanced lung cancer can be fitted quite accurately by an exponential distribution. Also, since log (observed mean survival) is approximately normal with variance 1/No. of deaths one can readily apply formula (7) to exponential data by setting $\sigma = 1$ and by defining σ_δ as in the earlier section on analysis of the exponential.

In this example, $N = 200$ and $N_H = 88$, the number of patients not deaths since it is reasonable to assume all patients in the historical group will have died before the results of the new trial are analyzed. Now, suppose one sets the bias of the mean survival in the historical controls to have a standard deviation of 5%. This is achieved by setting $\sigma_\delta = 0.56$. Then applying formula (7) we obtain

$$N_R = \tfrac{1}{2}\left(200 - \frac{88}{1 + (0.05)^2 \times 88}\right) = 63.9.$$

The above results indicate that the separate use of tumor response and survival data from the historical controls lead to slightly different sample size requirements for the randomized control group. Since patient survival might be considered the more reliable and important of the two measures one could allow survival results to predominate in the choice of N_R. In this case N_R is close to $N/3 = 67$ so that a convenient practical solution would be to randomize patients to Cytoxan + CCNU: Cytoxan + CCNU + Prednisone in the ratio of 1:2 in the new study.

The calculation of N_R is quite sensitive to the value of σ_δ. For instance, if for tumor response one chooses $\sigma_\delta = 5\%$ instead of 3% then N_R becomes 77 instead of 68. Similarly, if for survival one chooses $\sigma_\delta = 0.1$ instead of 0.05 then N_R becomes 77 instead of 64. As mentioned earlier, it is difficult to provide specific recommendations as to the value of σ_δ.

If $\sigma_\delta = 0$ then there is no bias in the historical data and

$$N_R = \max\left(0, \frac{N - N_H}{2}\right). \quad \text{As } \sigma_\delta \to \infty,\ N_R \to \frac{N}{2}$$

indicating that the historical data is considered worthless. The choice of σ_δ between these two extremes expresses the degree of trust one has in the historical data. An experimenter may have complete trust in his historical data (i.e. $\sigma_\delta = 0$ as far as he is concerned) but in practice he should choose a value of σ_δ which expresses the more cautious, sceptical views of the scientific community for which the results of the trial are intended.

SUMMARY

In many clinical trials the objective is to compare a new treatment with a standard control treatment, the design being to randomize equal numbers of patients onto the two treatments. However, there often exist acceptable historical data on the control treatment and this paper describes procedures for incorporating such historical controls into both the design and analysis of a randomized trial.

The statistical model supposes that treatment evaluation consists of a single quantitative measure for each patient and the objective of the trial is to estimate the true difference in treatment means for this measure. In general, historical controls cannot be considered as reliable as randomized controls and this leads one

to expect some bias in the historical data. This bias cannot be determined, even as regards its direction, and in the statistical model it is defined as a random variable with zero mean and variance to be specified. In practice, one might choose several values for this variance to represent varying degrees of mistrust, i.e. potential bias, in the historical data.

As regards analysis, the best estimate of the control treatment mean is a weighted average of the means for the randomized and historical controls. This leads to a more accurate comparison with the new treatment than the use of randomized controls alone. In the design of a randomized trial the presence of historical data enables one to enter a reduced proportion of patients into a randomized control group, the precise amount of this reduction depending on the size of the historical data and also its potential bias. Examples from actual sequences of clinical trials run by the Eastern Co-operative Oncology Group illustrate the practical use of the methods.

In conclusion, it is current practice in clinical trials to rely exclusively on either randomized controls or historical controls, but not both. The methods described in this paper provide an objective, quantitative approach for the combination of these two sources of control data and this should lead to a more efficient use of patients in the execution of clinical trials.

Acknowledgements—The author undertook most of this research while at the Statistical Laboratory, State University of New York at Buffalo, with support from National Cancer Institute grant CA-10810. The author is indebted to Dr. Marvin Zelen for invaluable advice on the statistical model and also to Dr. William Costello and Miss Sheila Gore for helpful comments. Thanks are expressed to the Eastern Co-operative Oncology Group for permission to use their data.

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